

Windows 8 and HTML5

Windows 8 will allow developers to create applications using web standard technologies such as HTML, CSS, and JavaScript

Firefox 5 released

Firefox 5, the first version of Firefox to be released under the new rapid release process is now available with support for CSS animations

Developer corner



Kshitij Sobti

kshtij.sobti@thinkdigit.com

Previously even simple animations in HTML required complex JavaScript code, and most complex animations needed Flash. Of course JavaScript is still incredibly useful if there has to be any form of interactivity, but new CSS3 features take a lot of the pain away. Let's look at two CSS3 modules – Transitions and Animation.

CSS transitions

The “:hover” selector in CSS already made it possible to make a page layout dynamic without requiring any JavaScript. Essentially, the “:hover” selector can be used to define the CSS styling of an element when a mouse hovers over it. It can be used to change the font size, the font / background colour, the border size / colour / type etc.

Try this, create an HTML file with just the following in it:

```
<!DOCTYPE html>
<html>
<head>
<title>Playing with CSS3</title>
<style type="text/css">
#div1 {
background-color: red;
}

#div2 {
border: solid thin black;
}

#div1:hover {
background-color: green;
}

#div2:hover {
border: dashed thick black;
```

```
}
#div3:hover {
font-size: 28pt;
}
</style></head>
<body>
<div id="div1">Background
colour sample</div>
<div id="div2">Border sample</div>
<div id="div3">Font sample</div>
</body>
</html>
```

If you save this in an HTML file (or use the files provided on this month's DVD, in the Deploy section), you'll see three rows. The first row is written in black on a red background, the second has a border, and the final one is just plain text. On moving your mouse over any row the formatting changes. The first row's background becomes green, the second row's border style changes and the third row's font size increases. You may notice that this change is rather jerky.

Change the CSS style for “div1” in the original file to:

```
#div1 {
background-color: red;
-moz-transition-duration: 3s;
-webkit-transition-duration: 3s;
-o-transition-duration: 3s;
transition-duration: 3s;
}
```

Now, when you hover your mouse over the first line, the colour will smoothly go from red to green. You'll notice we have “transition-duration” multiple times with different prefixes. To know why this is

CSS3 has made the web a much more dynamic place with support for transitions and animations. Let's take a look at these two new features.

needed, see the box *Vendor prefixes in CSS* on Page 66.

In this example, all we gave was the duration of the transition, i.e. three seconds, and the browser automatically animated all changed properties – those that are possible to animate obviously. A large number of CSS properties can be included in this transition, and it's even possible to specify which properties to change, and durations and easing functions for each.

Let's look at another small example. Here, on mouse hover we'll change the font and background colour, and add a shadow. We'll animate the background colour change over two seconds and the shadow over 500 milliseconds, but we won't animate the font colour. Here's the relevant part of the code (without the HTML structure):

```
HTML:
<div id="div1">Just a sample sentence</div>

CSS:
#div1 {
background-color: rgb(204, 204, 204);
transition-property: box-shadow, background-color;
transition-duration: 500ms, 2s;
}

#div1:hover {
box-shadow: 3px 3px 4px 1px black;
color: rgb(255, 255, 255);
background-color: rgb(102, 102, 102);
}
```

Please take note that this code as it is won't work! You'll need to add a prefix for each browser you want to use this on. So for Chrome, copy the “transition-property” and “transition-duration” lines and add “-webkit-” like before. We've left them out in this example and the ones that will follow to avoid redundancy.